

to critically test the five approaches. Rather, their intent was to provide opinions of senior managers about how and why their organizations function as they do. To that end, *Maximum Leadership* does an excellent job of demystifying what the world's leading CEOs do all day and why.

Farkas and De Backer also claim it was not their intent to offer a "silver bullet" for business success. Yet the anecdotal evidence presented—which is based on over 12,000 pages of interview transcripts—suggests that using the "right" approach can lead to organizational effectiveness or success. Given the tremendous range of organizations covered in this book, *Maximum Leadership* should be of interest to senior managers around the world and in every kind of business.

The Digital Economy: Promise and Peril in the Age of Networked Intelligence

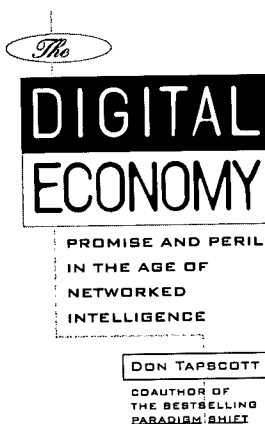
Reviewed by Joel P. Bowman, Western Michigan University

Book Review

At this point, it should be obvious to nearly everyone that we are in the throes of a revolution every bit as sweeping and significant as the Industrial Revolution of the late nineteenth and early twentieth centuries. Changes in technology and digital convergence—the merging of the computer, communications and information industries—are having a profound effect on virtually every aspect of our lives, and we have only recently begun to suspect how truly profound the effect will be. Digital information and "the digital networking of human intelligence" have become critical components of our economy.

Don Tapscott has been tracking and predicting cultural and economic changes fueled by technology since 1981. In *Office Automation* (Plenum Press, 1981), Tapscott predicted that computers would prove a double-edged sword, and this theme is echoed throughout *The Digital Economy*. Tapscott chronicles the increasing pressures on organizations to transform themselves through the total quality movement of the 1980s and the trend of business process reengineering of the 1990s. The reason such transformations typically fail is resistance to change: it is far easier to streamline processes and to cut costs by reducing the number of employees than it is to transform customer service, responsiveness and innovation.

Information technology (IT) makes the global economy possible. In a global economy, programmers and other knowledge workers in China, India or anywhere else in the world can become "virtual aliens" by telecommuting to work for U.S. corporations. The use of IT makes it possible for a wide variety of work to be done where the worker lives instead of in a central office or plant. In previous times, knowledge had to be transmitted by physical means: letters, reports, cash, checks, invoices, face-to-face meetings or telephone calls. Once it



By Don Tapscott, New York: McGraw-Hill, 1996. 342 pages, hard cover, \$24.95.

has been digitized, information can be transmitted at the speed of light from one location to another, thus eliminating many of the communication barriers caused by time and distance.

The new economy is also a knowledge economy. In the agrarian age, wealth was concentrated in land and land ownership. In the industrial age, wealth was increasingly concentrated in capital and the tools of production. In the information age, wealth is increasingly being concentrated in knowledge. According to Tapscott, nearly 60 percent of all American workers are knowledge workers, and eight out of ten new jobs are in information-intensive sectors of the economy. In such an economy, "Innovation, rather than access to resources, plant, and capital, is what counts most." In such an economy, it is not enough for organizations to learn how to play the game better; they must learn how to play a new game, a game in which the rules are constantly changing.

Tapscott identifies twelve overlapping themes in the new economy, including knowledge, digitization, disintermediation and discordance. The new economy will be reshaped by "leveraged intellect" and based on brain rather than brawn. An organization's most important assets are contained in the crania of its employees, who will increasingly be connected by digital networks. E-mail is just the beginning of the human collaboration that will become the driving force in the information age. Because knowledge and networking are global, organizations will be able to hire virtual employees who are present electronically rather than physically, and consumers will be able to shop in virtual malls.

"Middleman" functions—both middle management in organizations and agents of all varieties (insurance and real estate, for example)—are already being replaced by digital networks. Intermediates are increasingly unnecessary as critical information comes on line and owners and buyers become wired. Tapscott points out that electronic media have already become the dominant sector of the economy. The new rule is, "If you don't make it obsolete, someone else will." The life-cycle of individual products has fallen dramatically in recent years, and companies need to dramatically reduce the time spent developing and testing new products to keep pace.

The gap between producers and consumers is blurring as consumers are increasingly involved in the process of production. Consumers are increasingly demanding products designed and built to their specific needs. At the same time, product life cycles are "cratering." In 1990, automobiles took six years from concept to production. Today they take two. Currently, most of Hewlett-Packard's revenues come from products that didn't exist a year ago. Electronic Data Interchange (EDI) is "just the first splash in a tidal wave of electronic commerce" changing the relationships between companies.

In spite of the promise of the digital age, this technological revolution is producing every bit as much social trauma and conflict as was caused by the Industrial Revolution at the turn of the last century. The new economy is creating highly paid, high-value jobs, but there is little mobility between the old and the new. In the new economy, only those with "access to the new infrastructure can participate fully in social and commercial life."

In similar fashion, Tapscott identifies seven technological shifts driving the new economy, seven themes of internetworked government, and six educational themes influencing learning in the digital economy. Tapscott chronicles the ways in which work and learning are increasingly becoming the same thing and the shifting of teaching and learning away from formal schools and universities to the private sector. "Employee education is not growing 100 percent faster than academia, but 100 times—or 10,000 percent faster."

Everywhere we look, we can see signs that these themes are more than predictions and that they will have profound and far-reaching effects. No organization will long survive without becoming what Peter Senge has called a "learning organization." As learning becomes increasingly integrated with work, the private sector will assume greater responsibility for ensuring lifelong learning. In the old economy, basic competencies were relatively stable. In the new economy, workers must have up-to-the-minute knowledge to do their jobs, and knowledge is doubling every 18 months. According to Tapscott, "Nobody knows what a career plan is anymore."

In the new economy, an organization can compete only to the degree that it can learn more quickly than its competitors. Organizations need to overcome their "learning disabilities" by encouraging the networking of individual and team learning into an organizational consciousness encompassing the enterprise as a whole. Leadership based on the concept of command and control was sufficient in the old economy, but in the new economy, leadership will have to come from a collective, virtual force, with powers flowing from a jointly created and shared vision. The cultural leaders of the past relied on personal charisma to sell their individual visions. Leadership in the new economy will depend on empowered teams that can provide leadership for learning. A "sell-down" vision will no longer be sufficient; only a shared vision will work.

Although he is generally optimistic, Tapscott does not underestimate the difficulties we will face as every industry and nearly every job changes. Many enterprises today are overmanaged and underled, which inhibits the acquisition and dissemination of the knowledge required for economic survival. Twenty years ago, there were only 50,000 computers in the world; that many are now being installed daily. Those without knowledge and technology will be increasingly marginalized and we currently have no leadership for societal transformation.

The Digital Economy is an important book. Like other books that attempt to extrapolate the future based on the past and present, it doubtless misses the mark in some respects. Nevertheless, Tapscott has provided a wealth of support for his observations, conclusions and predictions. In spite of some redundancy from section to section, this is a powerful, highly readable book and one that deserves to be read by everyone who hopes to remain employed in the new, knowledge-based economy.

Copyright of *Academy of Management Executive* is the property of Academy of Management and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.